OPEN SOURCE / LOCAL BIOINFORMATICS / AGENT ASSISTANCE

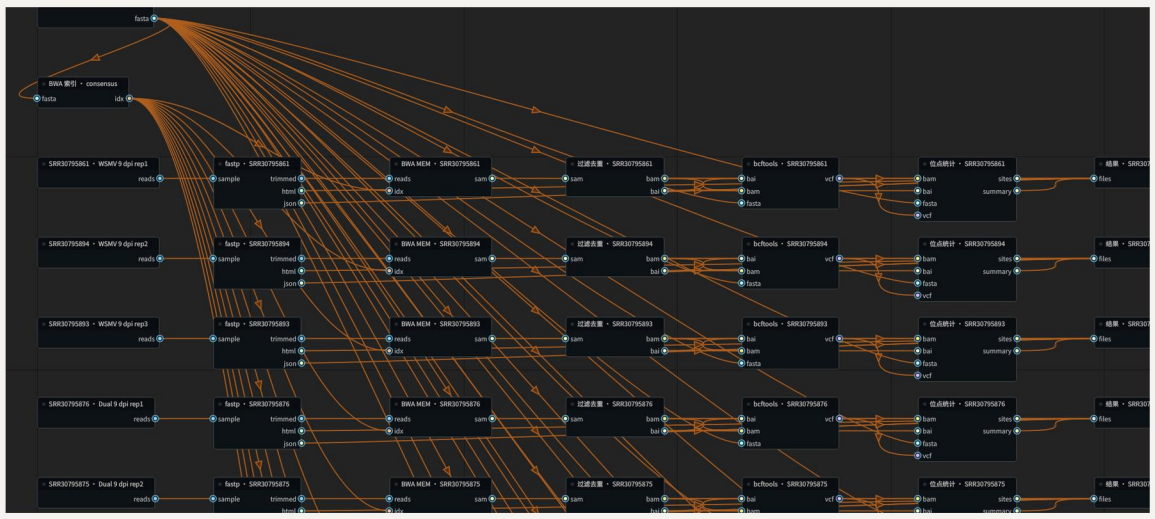
OpenBioWorkstation

Local workflows and agent collaboration

Visual bioinformatics with NodeGraphQt, Snakemake and MCP

NodeGraphQt / Snakemake / MCP

01 INPUT → TOOLS → OUTPUT



Actual WSMV workflow; includes case-specific nodes.
The original interface is preserved in the screenshot.

02 INSPECTION AND EXECUTION

Port format checks

```
[界面] 已拒绝连线 fastp · SRR30795861.json → 位点统计 · SRR30795861.bam：输出为 JSON，输入需要 BAM
```

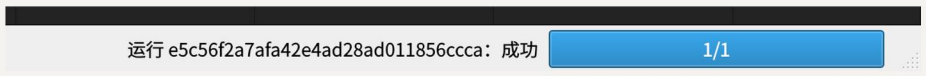
Reject incompatible formats

Execution states



Follow node execution states

Reruns and cache



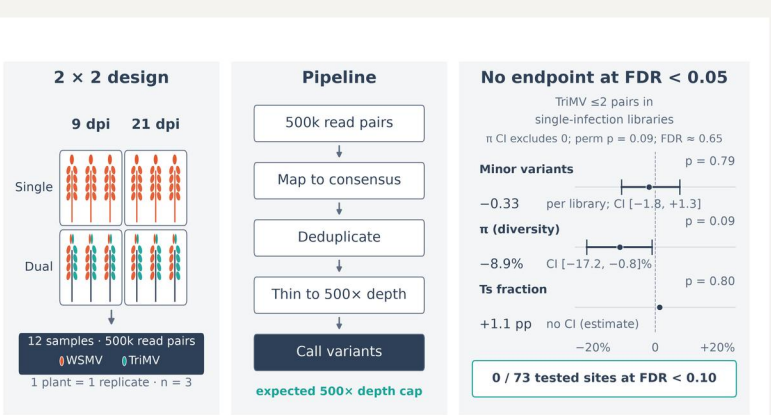
Partial runs and cached skips

File previews



Inspect bounded records

03 RESEARCH CASE: WSMV × TriMV



12 RNA-seq libraries

Existing application case

0/73 sites at FDR < 0.10

No site met the specified threshold

No significant association

No significant association detected

No equivalence claim

A negative result is not equivalence

04 WORK WITH AN AGENT

Question → agent-built workflow → dry-run → execute → inspect

Researcher oversight with agent-assisted workflow operations

Agent tutorial: docs/agent-workflow.en.md

Research examples: promo/research/

Linux, single machine. MCP stdio has been exercised.

Single machine; stdio tested; validate each agent client separately.

github.com/ArchitektSpeer/openbioworkstation

GPL-3.0